Conditional Probability

Mingxun Zhou (mingxunz@ust.hk)

COMP2711 Discrete Mathematical Tools

1 Conditional probability

A mother has two children. What is the probability that the younger child is a boy? The answer depends on what we know.

Assume each child is independently equally likely to be a boy or a girl. The four outcomes BB, BG, GB, GG are equally likely; the first letter describes the older child.

Given information	Remaining outcomes	Younger child is a boy
Older child is a boy	BB, BG	$1/2$
At least one child is a boy	BB, BG, GB	$2/3$

The condition changes which outcomes we consider.

Follow-up: A boy born on Wednesday.

A mother has two children. Given that at least one is a boy born on Wednesday, what is the probability that the other child is a boy? Here the target is that both children are boys.

Assume independent children, equally likely boy/girl, and independent, equally likely birth weekdays. Each child has $2 \cdot 7 = 14$ equally likely possibilities. Record the older child first, giving 14^2 outcomes.

Solution. Count outcomes satisfying the condition. Exclude those where neither child is a boy born on Wednesday:

$$14^2 - 13^2 = 27.$$

For both children to be boys, there are 7^2 weekday pairs. Exclude the 6^2 pairs where neither boy was born on Wednesday:

$$7^2 - 6^2 = 13.$$

The required probability is therefore $13/27$.

This conditions on a family having at least one boy born on Wednesday. It does not describe selecting one child randomly and observing that he is a boy born on Wednesday; under that procedure, the other child's probability of being a boy is $1/2$. How the information is obtained matters.

Notation and probability rules

An event is a subset of the sample space S. We write E and F for $E \cap F$, E or F for $E \cup F$, and $\overline{E}$ for the complement of E. Probabilities are nonnegative, $\Pr[S] = 1$, and probabilities of disjoint events add. In particular,

$$\Pr[\overline{E}] = 1 - \Pr[E], \qquad \Pr[E \text{ or } F] = \Pr[E] + \Pr[F] - \Pr[E \text{ and } F].$$

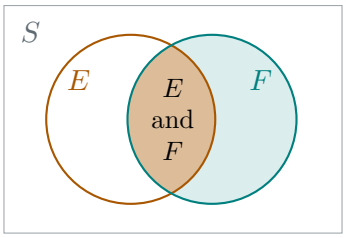
For a finite sample space with equally likely outcomes, $\Pr[E] = |E|/|S|$.

MUST UNDERSTAND Conditional probability

For $\Pr[F] > 0$, define

$$\Pr[E \mid F] = \frac{\Pr[E \text{ and } F]}{\Pr[F]}.$$

This is the probability of E given F: divide the probability that both events occur by the probability of F.



Keep only outcomes in F. The desired outcomes are those also in E. Dividing by $\Pr[F]$ makes the total probability within F equal to one. In the classical probability model, the remaining outcomes are equally likely, so

$$\Pr[E \mid F] = \frac{|E \cap F|}{|F|}.$$

For outcomes that are not equally likely, use probabilities, not counts. Conditioning on an event of probability zero is not defined by this formula.

Example: Two dice.

Roll two fair dice independently. Each ordered pair (i, j), with $1 \le i, j \le 6$, has probability $1/36$. Let E mean the sum is even and F mean at least one die shows 6.

There are $6 + 6 - 1 = 11$ outcomes in F: six with the first die 6 and six with the second die 6, with $(6, 6)$ counted twice. Of these, the even-sum outcomes are

$$(2, 6),\ (4, 6),\ (6, 6),\ (6, 4),\ (6, 2).$$

Thus

$$\Pr[E \mid F] = \frac{|E \cap F|}{|F|} = \frac{5}{11}.$$

For the reverse condition, an even sum requires two odd numbers or two even numbers: $|E| = 3^2 + 3^2 = 18$. Therefore

$$\Pr[F \mid E] = \frac{|E \cap F|}{|E|} = \frac{5}{18}.$$

Reversing the condition changes the denominator.

Interpreting the condition

Conditioning need not describe causation. Pr[alarm sounds | fire] asks: given a fire, how likely is the alarm to sound? A fire can cause an alarm. By contrast, Pr[fire | alarm sounds] asks: given an alarm, how likely is there to be a fire? The alarm provides evidence; it does not cause the fire. A conditional probability alone does not establish causation.

Conditioning does not specify time order. Pr[rain tomorrow | rain today] is a prediction about the future. Pr[rain yesterday | rain today] is inference about the past. Both use the same observation, rain today.

Example: A five-card hand.

Draw five cards uniformly without replacement from a standard 52-card deck. Let E mean exactly one ace and F mean at least one ace. Since $E \subseteq F$, knowing E guarantees F: $\Pr[F \mid E] = 1$.

For the reverse condition, count hands:

$$|E| = \binom{4}{1}\binom{48}{4}, \qquad |F| = \binom{52}{5} - \binom{48}{5}.$$

The second expression subtracts hands drawn entirely from the 48 non-aces. Since $E \cap F = E$,

$$\Pr[E \mid F] = \frac{\binom{4}{1}\binom{48}{4}}{\binom{52}{5} - \binom{48}{5}}.$$

Follow-up. What is the probability of at least one ace, given at least one king? The denominator counts hands with a king. For the numerator, subtract hands with no aces or no kings, using inclusion–exclusion:

$$\Pr[\text{at least one ace} \mid \text{at least one king}] = \frac{\binom{52}{5} - 2\binom{48}{5} + \binom{44}{5}}{\binom{52}{5} - \binom{48}{5}}.$$

The last term restores hands containing neither an ace nor a king, which were subtracted twice.

2 The multiplication rule

MUST UNDERSTAND Multiplication rule

Rearrange the conditional-probability definition:

$$\Pr[E \text{ and } F] = \Pr[E]\Pr[F \mid E] \qquad (\Pr[E] > 0).$$

First require E; then use the probability of F given E. The order can also be reversed:

$$\Pr[E \text{ and } F] = \Pr[F]\Pr[E \mid F] \qquad (\Pr[F] > 0).$$

This rule does not require independence.

Example: Consecutive aces.

Draw cards uniformly without replacement. Let A_i mean draw i is an ace. After drawing one ace, there are three aces among 51 remaining cards. Hence

$$\Pr[A_1 \text{ and } A_2] = \frac{4}{52} \cdot \frac{3}{51} = \frac{1}{221}.$$

Direct counting gives the same answer: $\binom{4}{2}/\binom{52}{2} = 1/221$. For three aces, the next factor is $2/50$:

$$\Pr[A_1 \text{ and } A_2 \text{ and } A_3] = \frac{4}{52} \cdot \frac{3}{51} \cdot \frac{2}{50} = \frac{1}{5525}.$$

General chain rule

For events $E_1, \ldots, E_n$,

$$\Pr\left[\bigcap_{i=1}^{n} E_i\right] = \Pr[E_1] \prod_{i=2}^{n} \Pr\left[E_i \,\middle|\, \bigcap_{j=1}^{i-1} E_j\right],$$

provided every conditioning event has positive probability. Each factor conditions on all preceding events, not just the previous one. If $\Pr[E_1 \text{ and } \cdots \text{ and } E_j] = 0$ for some j, then the probability that all n events occur is already zero.

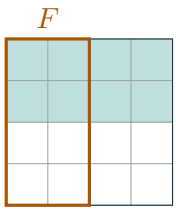
3 Independence and complements

MUST UNDERSTAND Independent events

Events E, F are independent when

$$\Pr[E \text{ and } F] = \Pr[E] \Pr[F].$$

For $\Pr[F] > 0$, this is equivalent to $\Pr[E \mid F] = \Pr[E]$: learning F does not change the probability of E. The product definition also applies when an event has probability zero.



E: upper two rows

$\Pr[E] = 8/16 = 1/2$

$\Pr[E \mid F] = 4/8 = 1/2$

Each square is equally likely. Restricting to the left two columns leaves the proportion in E unchanged. The lower two rows represent $\overline{E}$; their proportion is also $1/2$, both before and after conditioning.

Example: An ace and a heart.

Draw one card uniformly. Let A mean ace and H mean heart. Among the 13 hearts, exactly one is an ace, so

$$\Pr[A \mid H] = \frac{1}{13} = \frac{4}{52} = \Pr[A].$$

Equivalently, $\Pr[A \text{ and } H] = 1/52 = (4/52)(13/52)$. Events can be independent even when they describe the same draw.

Replacement and disjoint events

With replacement and reshuffling, the first and second draws are independent. Without replacement, learning that the first card is an ace changes the second draw's ace probability from $4/52$ to $3/51$. Before seeing the first card, the second card is still equally likely to be any of the 52 cards.

Disjoint events of positive probability are not independent: $\Pr[E \text{ and } F] = 0 < \Pr[E]\Pr[F]$. For one card, heart and spade are disjoint. Knowing the card is a spade changes the probability of a heart from $1/4$ to zero.

Independence and complements

If E, F are independent, complementing either or both preserves independence. For example,

$$\begin{aligned}\Pr[\overline{E} \text{ and } F] &= \Pr[F] - \Pr[E \text{ and } F] \\ &= \Pr[F] - \Pr[E]\Pr[F] \\ &= (1 - \Pr[E])\Pr[F] = \Pr[\overline{E}]\Pr[F].\end{aligned}$$

Interchanging E, F proves independence of $E, \overline{F}$. Applying the same result again proves independence of $\overline{E}, \overline{F}$.

Pairwise and mutual independence

Three events A, B, C are pairwise independent when each pair satisfies the product rule. They are mutually independent when these three pair equations hold and also

$$\Pr[A \text{ and } B \text{ and } C] = \Pr[A]\Pr[B]\Pr[C].$$

For n events, mutual independence requires the product equation for every subset of two or more events, not just for the full intersection.

Example: Two coin flips.

Flip a fair coin twice independently. Let A mean the first flip is heads, B the second flip is heads, and C exactly one flip is heads.

Outcome	A	B	C
HH	Yes	Yes	No
HT	Yes	No	Yes
TH	No	Yes	Yes
TT	No	No	No

Each event has probability 1/2, and each pair intersection has probability 1/4. They are pairwise independent. But their triple intersection is empty, so its probability is 0, not $(1/2)^3$.

4 Total probability

Example: A medical test.

Suppose 1% of a population has a disease. A person with the disease tests positive with probability 95%; a person without it tests positive with probability 10%. Select a person randomly. What is the probability of a positive result?

Let D mean the person has the disease and T mean the test is positive. Split positive results into two disjoint cases: people with the disease and people without it. Then

$$\begin{aligned}\Pr[T] &= \Pr[T \mid D]\Pr[D] + \Pr[T \mid \overline{D}]\Pr[\overline{D}] \\ &= 0.95(0.01) + 0.10(0.99) = 0.1085.\end{aligned}$$

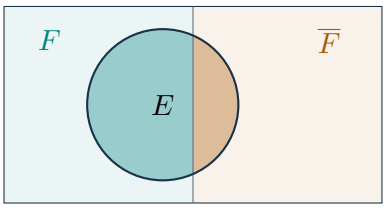
The answer is 10.85%. This is $\Pr[T]$, not $\Pr[D \mid T]$. The latter is the question studied in the next lecture on Bayes' rule.

MUST UNDERSTAND Total probability

Let $F_1, \ldots, F_m$ be disjoint events covering the sample space, with $\Pr[F_i] > 0$. Then

$$\Pr[E] = \sum_{i=1}^{m} \Pr[E \mid F_i]\Pr[F_i].$$

The events $E \cap F_i$ split E into disjoint pieces. Add their probabilities, using the multiplication rule for each piece.



For two cases, this is $\Pr[E] = \Pr[E \mid F]\Pr[F] + \Pr[E \mid \overline{F}]\Pr[\overline{F}]$. The diagram shows how the pieces join; their drawn areas need not represent their probabilities.

Example: Three bags.

Choose bag A, B, C with probabilities $0.5, 0.3, 0.2$, respectively. Each bag has four balls: A has one red, B two red, and C three red. After choosing a bag, draw one ball uniformly. Let F_A, F_B, F_C identify the chosen bag and R mean a red ball. Then

$$\begin{aligned}\Pr[R] &= \Pr[R \mid F_A]\Pr[F_A] + \Pr[R \mid F_B]\Pr[F_B] + \Pr[R \mid F_C]\Pr[F_C] \\ &= \frac{1}{4}(0.5) + \frac{2}{4}(0.3) + \frac{3}{4}(0.2) = 0.425.\end{aligned}$$

Optional: Monty Hall

A prize is placed uniformly behind one of three doors. You choose door 1. The host opens door 3, revealing no prize, and offers a switch to door 2. Should you switch? The answer depends on how the host selected a door. Let F denote the observed event: the host opens door 3 and it is empty.

Case 1: The host knows. The host always opens an unchosen empty door, chooses uniformly if both are empty, and always offers a switch.

Case 2: The host chooses randomly. The host chooses door 2 or 3 uniformly, independently of the prize. This can reveal the prize, but on this occasion the opened door is empty. After an empty reveal, a switch is offered.

Prize location	Prior probability	$\Pr[F \mid \text{location}]$, case 1	$\Pr[F \mid \text{location}]$, case 2
Door 1	1/3	1/2	1/2
Door 2	1/3	1	1/2
Door 3	1/3	0	0

In case 1, the joint probabilities of F with prize locations 1 and 2 are $1/6$ and $1/3$. Thus

$$\Pr[\text{prize at door 2} \mid F] = \frac{1/3}{1/6 + 1/3} = \frac{2}{3}.$$

In case 2, the two joint probabilities are both $1/6$, so the answer is $(1/6)/(1/6 + 1/6) = 1/2$. The same observation gives different answers under different rules. In case 1, the assumption that the host chooses uniformly between two empty doors matters because we condition on a specific door being opened.

Extension: Conditional independence

Independence can change when we receive information.

For $\Pr[A] > 0$, events B_1, B_2 are **conditionally independent given A** when

$$\Pr[B_1 \text{ and } B_2 \mid A] = \Pr[B_1 \mid A] \Pr[B_2 \mid A].$$

In words, B_1 and B_2 are independent after conditioning on A. For events $B_1, \ldots, B_n$, conditional mutual independence given A means that for every nonempty index set $I \subseteq \{1, \ldots, n\}$,

$$\Pr\left[\bigcap_{i \in I} B_i \,\middle|\, A\right] = \prod_{i \in I} \Pr[B_i \mid A].$$

Checking pairs alone is not sufficient for three or more events.

For example, to update the probability of A after observing both B_1 and B_2, Bayes' rule uses the probability $\Pr[B_1 \text{ and } B_2 \mid A]$:

$$\Pr[A \mid B_1 \text{ and } B_2] = \frac{\Pr[B_1 \text{ and } B_2 \mid A] \Pr[A]}{\Pr[B_1 \text{ and } B_2]}, \qquad \Pr[B_1 \text{ and } B_2] > 0.$$

Conditional independence lets us replace $\Pr[B_1 \text{ and } B_2 \mid A]$ by $\Pr[B_1 \mid A] \Pr[B_2 \mid A]$. Separate observations need not be conditionally independent.

Independence can disappear after conditioning

Flip a fair coin twice independently. Let H_1, H_2 mean heads on the first and second flips. These events are independent: their probabilities are $1/2$, and their joint probability is $1/4$.

Now condition on T: at least one flip is tails. The remaining outcomes are HT, TH, TT, equally likely. Therefore

$$\Pr[H_1 \mid T] = \Pr[H_2 \mid T] = \frac{1}{3}, \qquad \Pr[H_1 \text{ and } H_2 \mid T] = 0 \neq \frac{1}{9}.$$

They are dependent given T. Within the condition, learning that the first flip is heads forces the second to be tails.

The conditioning event matters

Conditional independence given A implies neither unconditional independence nor independence given $\overline{A}$. One die example shows both distinctions. Roll a fair die and define

$$A = \{1, 2, 3, 4\}, \qquad B_1 = \{1, 2, 5\}, \qquad B_2 = \{1, 3, 5\}.$$

Outcome	In A?	In B_1?	In B_2?
1	Yes	Yes	Yes
2	Yes	Yes	No
3	Yes	No	Yes
4	Yes	No	No
5	No	Yes	Yes
6	No	No	No

Given A, keep the first four rows. Each event occurs in two rows and both occur in one row:

$$\Pr[B_1 \text{ and } B_2 \mid A] = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} = \Pr[B_1 \mid A] \Pr[B_2 \mid A].$$

Thus they are independent given A. Without conditioning, each event occurs in three rows and both in two:

$$\Pr[B_1 \text{ and } B_2] = \frac{2}{6} \neq \frac{3}{6} \cdot \frac{3}{6} = \Pr[B_1] \Pr[B_2].$$

They are not independent without the condition. Given $\overline{A}$, keep only rows 5 and 6. The two events now occur together:

$$\Pr[B_1 \text{ and } B_2 \mid \overline{A}] = \frac{1}{2} \neq \frac{1}{4} = \Pr[B_1 \mid \overline{A}] \Pr[B_2 \mid \overline{A}].$$

Complementing one of two independent events preserves independence. Changing the conditioning event does not.

Problem Set

Full solutions follow the problem set.

1. **One die.** Roll a fair die. Let $E = \{2, 4, 6\}$ and $F = \{4, 5, 6\}$. Find $\Pr[E \mid F]$ and $\Pr[F \mid E]$. List the remaining outcomes in each case.

2. **Reasoning backward.** A bag has three red and two blue balls. Draw three uniformly without replacement. Given that the third ball is red, what is the probability that the first two have different colors?

3. **Test independence.** Roll a fair die. Let

$$E = \{2, 4, 6\}, \qquad F = \{4, 5, 6\}, \qquad G = \{1, 2, 3, 4\}.$$

   Are E, F independent? Are E, G independent? Justify your answers.

   Let $H \subseteq \{1, 2, 3, 4, 5, 6\}$.

   (a) If H is independent of E, what values can $|H|$ take? Give an example for each possible size.

   (b) Can a nonempty proper subset H be independent of both E and G? Give an example or prove that none exists.

4. **Unknown dependence.** You know only $\Pr[E] = 0.6$ and $\Pr[F] = 0.5$. What are the smallest and largest possible values of $\Pr[E \mid F]$? What if E, F are independent?

5. **Three flips.** Flip a fair coin three times independently. Let A mean the first flip is heads, B the second flip is heads, and C all three results are the same. Are A, B, C pairwise independent? Mutually independent?

6. **Same bag, two draws.** A fair coin selects bag A (three red, one blue) or bag B (one red, three blue). Draw twice uniformly from that same bag, replacing and mixing after the first draw. Let R_i mean draw i is red. Find $\Pr[R_1 \text{ and } R_2]$ and $\Pr[R_2 \mid R_1]$. Are R_1, R_2 independent?

Solutions

Solution 1: One die.

Given F, the remaining outcomes are $\{4, 5, 6\}$. Given E, the remaining outcomes are $\{2, 4, 6\}$. In each case, the favorable outcomes are $\{4, 6\}$, so

$$\Pr[E \mid F] = \frac{2}{3}, \qquad \Pr[F \mid E] = \frac{2}{3}.$$

Equal conditional probabilities do not imply independence.

Solution 2: Reasoning backward.

Let D mean the first two colors differ, and R_3 mean the third ball is red. Each position is equally likely to contain any of the five balls, so $\Pr[R_3] = 3/5$. The qualifying color orders are red–blue–red and blue–red–red:

$$\Pr[D \text{ and } R_3] = \frac{3}{5} \cdot \frac{2}{4} \cdot \frac{2}{3} + \frac{2}{5} \cdot \frac{3}{4} \cdot \frac{2}{3} = \frac{2}{5}.$$

Therefore

$$\Pr[D \mid R_3] = \frac{2/5}{3/5} = \frac{2}{3}.$$

Alternative. Reserve a red ball for the third position. The remaining four balls are two red and two blue. The first two draws have different colors with probability

$$\frac{2}{4} \cdot \frac{2}{3} + \frac{2}{4} \cdot \frac{2}{3} = \frac{2}{3}.$$

Information about a later draw can change probabilities for earlier draws.

Solution 3: Test independence.

Check each product:

$$\Pr[E \text{ and } F] = \frac{2}{6} \neq \frac{3}{6} \cdot \frac{3}{6}, \qquad \Pr[E \text{ and } G] = \frac{2}{6} = \frac{3}{6} \cdot \frac{4}{6}.$$

Thus E, F are dependent, whereas E, G are independent. In Exercise 1, $\Pr[E \mid F] = \Pr[F \mid E] = 2/3$, but independence would require $\Pr[E \mid F] = \Pr[E] = 1/2$.

Follow-up (a). Independence of H and E requires

$$\frac{|H \cap E|}{6} = \frac{|H|}{6} \cdot \frac{1}{2}, \qquad \text{so} \qquad |H \cap E| = \frac{|H|}{2}.$$

Thus H must contain equally many even and odd outcomes. Its possible sizes are $0, 2, 4, 6$, with examples $\emptyset$, $\{1, 2\}$, $\{1, 2, 3, 4\}$, and $\{1, 2, 3, 4, 5, 6\}$.

Follow-up (b). Independence of H and G also requires

$$|H \cap G| = \frac{2}{3}|H|.$$

The left side is an integer, so $|H|$ must be divisible by 3. Part (a) requires $|H|$ to be even. Hence $|H|$ is divisible by 6: only 0 and 6 are possible. Neither gives a nonempty proper subset, so no such H exists. Here independence of both means independence of each event separately; mutual independence is not assumed.

Solution 4: Unknown dependence.

Write $t = \Pr[E \text{ and } F]$. The overlap is contained in F, so $t \le 0.5$. Also,

$$\Pr[E \text{ or } F] = 0.6 + 0.5 - t \le 1, \qquad \text{so } t \ge 0.1.$$

Divide by $\Pr[F] = 0.5$:

$$0.2 \le \Pr[E \mid F] \le 1.$$

Both endpoints are possible. In ten equally likely outcomes, take $E = \{1, 2, 3, 4, 5, 6\}$. Choosing $F = \{6, 7, 8, 9, 10\}$ gives overlap probability 0.1 and conditional probability 0.2; choosing $F = \{1, 2, 3, 4, 5\}$ gives conditional probability 1. If the events are independent, then $t = 0.6 \cdot 0.5 = 0.3$ and $\Pr[E \mid F] = 0.6$. Knowing $\Pr[E]$ and $\Pr[F]$ alone does not determine $\Pr[E \text{ and } F]$.

Solution 5: Three flips.

The eight strings are equally likely. $C = \{HHH, TTT\}$, so $\Pr[A] = \Pr[B] = 1/2$ and $\Pr[C] = 1/4$. The intersections have probabilities

$$\Pr[A \text{ and } B] = \frac{1}{4}, \qquad \Pr[A \text{ and } C] = \Pr[B \text{ and } C] = \frac{1}{8}.$$

Each equals the corresponding product, so the events are pairwise independent. Only HHH belongs to all three:

$$\Pr[A \text{ and } B \text{ and } C] = \frac{1}{8} \ne \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{16}.$$

They are not mutually independent.

Solution 6: Same bag, two draws.

Given the chosen bag, the draws are independent. Split into the two bag choices and use total probability:

$$\Pr[R_1 \text{ and } R_2] = \frac{1}{2}\left(\frac{3}{4}\right)^2 + \frac{1}{2}\left(\frac{1}{4}\right)^2 = \frac{5}{16}.$$

For either draw,

$$\Pr[R_i] = \frac{1}{2} \cdot \frac{3}{4} + \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{2}.$$

Thus

$$\Pr[R_2 \mid R_1] = \frac{5/16}{1/2} = \frac{5}{8}.$$

The events are not independent: $5/16 \ne (1/2)(1/2)$. Seeing red makes the red-rich bag more likely, increasing the chance of another red. Replacement keeps the bag's contents unchanged, but the first draw gives information about which bag was selected. The draws are independent given the bag, but not independent without that information.